

AutoGen + RAG for E-Commerce — Project Explanation

Overview

This project is the seventh entry in my Agentic AI series and a genuine crossover point with the RAG-focused notebooks built earlier using LangChain: it rebuilds the same retrieve-then-generate pattern entirely inside AutoGen's multi-agent architecture, using two independent ChromaDB-backed retrieval crews for an e-commerce product catalogue and customer order dataset.

What the Notebook Does

Product and order data from two CSV files are loaded into two separate in-memory ChromaDB collections, kept apart so each retrieval agent only ever searches its own domain. Two search functions are registered onto separate assistant and executor agents using AutoGen's native tool-calling decorators, `register_for_llm` and `register_for_execution`, which split responsibility for knowing a tool exists from actually being allowed to run it, a different design from single-node tool execution patterns used elsewhere in this series.

Each retrieval domain runs as its own small group chat using round-robin speaker selection, a third speaker-selection strategy alongside the automatic and custom state-machine approaches covered in an earlier notebook, appropriate here since a two-agent search-and-execute loop has an obviously fixed turn order. Both group chats are then run concurrently through AutoGen's asynchronous chat initiation function, a real efficiency gain over running them strictly one after another. Finally, a dedicated writer agent, explicitly instructed to answer only from the retrieved information and to admit when it does not know something, synthesises the final answer from both retrieval crews' combined findings, mirroring the same retrieval-then-generation split used throughout the RAG notebooks earlier in this portfolio.

Skills Demonstrated

- Building RAG retrieval entirely with AutoGen agents rather than LangChain chains
- AutoGen's native function-calling pattern across assistant and executor agents
- Concurrent, asynchronous multi-agent orchestration
- Separating retrieval agents from a dedicated answer-synthesis agent

This crossover between RAG grounding techniques and multi-agent orchestration is a direct rehearsal for the retrieval-and-synthesis agents in my dissertation's agentic intelligence platform.